

Subject: Macroeconomics | NCERT Class 12 Ch. 4 | UPSC & State PSC Core

30-Second Snapshot

- Developed by J.M. Keynes, this short-run model assumes **fixed price levels and interest rates**, meaning output is determined entirely by **Effective Demand**.
- Decisions depend on **ex ante (planned) values** rather than **ex post (realized) accounting figures**.
- The consumption function combines baseline **autonomous spending** with **induced consumption** driven by the Marginal Propensity to Consume (MPC).
- Equilibrium is achieved where planned national output matches planned aggregate demand, with **unintended inventory changes** acting as the self-adjusting market signal.
- The **Investment Multiplier** equals the inverse of the Marginal Propensity to Save ($1 / \text{MPS}$), meaning higher spending propensities produce greater output expansions.
- The **Paradox of Thrift** shows that collective attempts to save a higher share of income depress demand and reduce national income, leaving total savings unchanged or lower.

Aggregate Demand Parameters & Propensity Matrix

Macro Parameter	Operational Formula	Economic Definition	Baseline Boundary Values
Autonomous Consumption	C_{bar}	Non-income-dependent spending required for basic survival.	Strictly greater than 0
Marginal Propensity to Consume	$\text{MPC} = \frac{\text{Change in } C}{\text{Change in } Y}$	Fraction of an additional rupee of income directed toward consumption.	Strictly between 0 and 1
Marginal Propensity to Save	$\text{MPS} = \frac{\text{Change in } S}{\text{Change in } Y}$	Fraction of an additional rupee of income retained as savings.	$\text{MPS} = 1 - \text{MPC}$
Average Propensity to Consume	$\text{APC} = C / Y$	Total proportion of aggregate national income spent on consumption.	Falls as national income rises
Average Propensity to Save	$\text{APS} = S / Y$	Total proportion of aggregate national income retained as savings.	$\text{APC} + \text{APS} = 1$
Investment Multiplier	$k = 1 / \text{MPS} = 1 / (1 - \text{MPC})$	Magnitude of total income expansion resulting from a unit change in spending.	Always greater than or equal to 1

Adjustment Mechanics & Macroeconomic Output Gaps

- The Unplanned Inventory Balancing Mechanism:**
 - Output Exceeds Demand ($Y > AD$):** Unsold inventories pile up unexpectedly. In response, factories reduce production schedules and lay off workers until output contracts back to equilibrium.
 - Demand Exceeds Output ($AD > Y$):** Inventories are drawn down unexpectedly to satisfy orders. In response, factories hire labor and scale up production until output expands to equilibrium.
 - Equilibrium State:** Output equals aggregate demand, meaning unintended inventory adjustments drop to zero.
- The Deflationary Gap (Deficient Demand):**
 - Planned aggregate demand settles below the output capacity required for full employment.
 - Leaves productive capacity idle and causes involuntary unemployment without

automatic price corrections in the short run.

- Requires countercyclical public capital expenditures or tax reductions to bridge the gap.

- **The Inflationary Gap (Excess Demand):**

- Planned aggregate demand outstrips output capacity at full employment.
- Because physical production cannot exceed full resource limits, the excess demand drives up general price levels.

⚠ Common UPSC Exam Traps

- **Trap 1 (Ex Ante Plans vs. Ex Post Realities):** Macroeconomic equilibrium is established when ex ante planned aggregate demand matches ex ante planned output. It is not defined by ex post realized accounting equality, which always matches identically by definition.
- **Trap 2 (The Multiplier-MPS Relationship):** The investment multiplier is directly proportional to the Marginal Propensity to Consume (MPC), but inversely proportional to the Marginal Propensity to Save (MPS). A higher savings propensity reduces the multiplier.
- **Trap 3 (The Paradox of Thrift Confusion):** While saving more is financially prudent for an individual household, an economy-wide drive to save more lowers aggregate demand, cuts factory production, and reduces total national income.
- **Trap 4 (Equilibrium Does Not Equal Full Employment):** A Keynesian goods market equilibrium merely indicates that planned production matches planned demand. It frequently settles at an under-employment equilibrium with persistent joblessness.

🔗 High-Yield Facts Box

Parameter / Concept	Operational Benchmark	Policy & Governance Relevance
Effective Demand Principle	Output equals Aggregate Demand	Proves that deficient spending, not worker unwillingness, causes recessions.
Multiplier Identity	$k = 1 / (1 - MPC)$	Explains why targeted public capital investments produce broader economic gains.
Paradox of Thrift	Higher MPS reduces national income	Discourages across-the-board austerity measures during deep recessions.
Inventory Changes	Buffer deviations trigger output shifts	Critical high-frequency indicator used to track business cycles.
Deflationary Gap Solution	Increase public spending (G)	Justifies government fiscal deficit expansion during economic downturns.

📝 Prelims Practice Challenge

Q1. Consider the following statements regarding the Keynesian model of income determination:

1. An increase in the Marginal Propensity to Save leads to a higher value for the investment multiplier.
2. The economy reaches equilibrium only when unintended inventory accumulation or depletion is reduced to zero.
3. Macroeconomic equilibrium in the goods market always guarantees the full employment of all available labor resources.

Which of the statements given above is/are correct?

- (A) 1 and 2 only
- (B) 2 only
- (C) 2 and 3 only
- (D) 1, 2, and 3

Answer: (B) 2 only

Explanation: Statement 1 is incorrect because the multiplier equals $1 / \text{MPS}$; a higher MPS lowers the multiplier. Statement 2 is correct because equilibrium requires planned output to match planned demand, which eliminates unplanned inventory changes. Statement 3 is incorrect because Keynesian equilibrium can and frequently does occur below full employment capacity.

Q2. Consider the following statements regarding the Paradox of Thrift:

1. It shows that an increase in the individual desire to save can lead to an overall decline or stagnation in aggregate national savings.
2. It occurs because higher savings rates expand aggregate consumer demand and encourage new factory production.

Which of the statements given above is/are correct?

- (A) 1 only
- (B) 2 only
- (C) Both 1 and 2
- (D) Neither 1 nor 2

Answer: (A) 1 only

Explanation: Statement 1 is correct because collective attempts to save reduce total consumer demand and depress national income, which prevents total savings from rising. Statement 2 is incorrect because an increased propensity to save reduces consumption expenditure, creating excess inventories and forcing output cuts.

 Mains Practice Question (10 Marks / 150 Words)

Question: "Explain the Keynesian concept of the Investment Multiplier. How does the Paradox of Thrift challenge traditional microeconomic assumptions regarding savings and capital accumulation?"

 Structured Answer Framework:

• **Introduction (25–30 words):**

Define the Investment Multiplier as the ratio of change in national income to an initial change in autonomous expenditure ($k = 1 / \text{MPS}$). Explain that initial injections trigger multiple rounds of induced consumer spending.

• **Body Arguments (80–90 words):**

- *The Multiplier Cycle:* An initial capital investment pays out wages and profits. Recipients spend a fraction of this income based on their Marginal Propensity to Consume, generating sequential rounds of income and employment.
- *Micro versus Macro Logic:* In microeconomics, individual saving is considered a virtue that funds personal wealth and provides investment capital.
- *The Aggregate Demand Fallacy:* When applied economy-wide during a slump, higher savings propensities (higher MPS) lower consumption, create unintended inventory buildups, force factories to cut production, and drag down national income.

• **Conclusion / Way Forward (25–30 words):**

Recognizing the Paradox of Thrift shows why governments must deploy countercyclical fiscal spending and support purchasing power during recessions, rather than relying on private thrift alone.